

1. THE CENTRAL FALSIFIABLE CLAIM

CORE THESIS

5. WHAT THE LEARNER CAN TEST (INTERACTIVE SIMULATOR)

6 PRESETS

“A fixed-size evolving state can process sequences whose duration grows without allocating a new memory slot for every token, but limited state capacity can cause interference and forgetting.”

A fixed-size state representation remains fixed with respect to sequence length T for a fixed model configuration ($O(d^2)$ matrix parameters), avoiding linear growth in cached-token memory ($O(T \cdot d)$ in Transformer KV caches). However, fixed representation size does not imply unlimited capacity or zero information loss: past facts are superposed, creating cross-talk noise and selective forgetting under sequence pressure.

2. RECURRENT MECHANISM & RETRIEVAL DECOMPOSITION

MATH SUBSTRATE

The educational substrate models an explicit 2D matrix state $S_t \in \mathbb{R}^{d \times d}$. Given normalized key-value pairs (k_t, v_t) , the state updates incrementally via:

$$S_t = \lambda S_{t-1} + \eta (v_t \otimes k_t^T) = \sum_{\tau=1}^t \lambda^{t-\tau} \eta (v_\tau k_\tau^T)$$

Probing with query vector q retrieves $\hat{v} = S_t q$, which decomposes analytically into signal plus cross-talk noise:

$$\hat{v} = \lambda^{t-\tau_{\text{target}}} \eta (k_{\text{target}}^T q) v_{\text{target}} + \sum_{\tau \neq \text{target}} \lambda^{t-\tau} \eta (k_\tau^T q) v_\tau$$

Signal-to-Noise Ratio: $\text{SNR}_{\text{dB}} = 10 \log_{10}(|s|^2 / (|n|^2 + \epsilon))$. When $T \leq d$ and keys form an orthonormal set ($k_\tau^T q = 0$ for $\tau \neq \text{target}$), noise is zero. But because $\mathbb{R}^d$ supports at most d mutually orthogonal directions, storing $T > d$ associations forces non-orthogonal superpositions that accumulate cross-talk noise.

5. WHAT THE LEARNER CAN TEST (INTERACTIVE SIMULATOR)

The interactive artifact provides an intentionally small educational associative-memory model inspired by recurrent state architectures, demonstrating 6 deterministic test regimes (Mulberry32 PRNG):

Clean Baseline (T=5, d=16) Orthonormal keys; noise = 0.000; SNR = +90 dB; 100% exact retrieval.	Capacity Bound (T=16, d=16) State approaches full rank; mild cross-talk begins (SNR = +6.9 dB).
Severe Overload (T=60, d=8) T >> d; cross-talk exceeds signal (SNR = -8.5 dB); interference failure.	Selective Decay (T=35, d=16, λ=0.92) Norm bounded; early fact decays by 94.1% (λ ³⁴); recency amnesia.
Key Collision (T=20, d=16, p=0.70) Correlated keys cause leakage even with relatively large state size.	Break-It Stress (T=250, d=8) Empirical phase-transition curve mapping accuracy vs T/d ratio.

6. PATHWAY FRONTIER CONNECTION: BDH & BDH-CQ

FRONTIER ROLE

Pathway researchers resolved recurrent representational bottlenecks in **Dragon Hatchling (BDH, arXiv:2509.26507)** by reformulating attention as dynamic synaptic plasticity over sparse particle graphs with Q=K Hebbian self-affinity and non-negative activations. **BDH-CQ (arXiv:2608.09888)** extends this to in-context learning: demonstrations update recurrent memory at inference time, and queries are solved iteratively in high-dimensional latent space without verbalizing text tokens.

Published Result: On public ARC-AGI-1 evaluation, a 150M BDH-CQ model achieved **29.5% pass@2 at \$0.0007 per task** (under 1/10th cent), establishing a reported cost-efficiency Pareto point.

7. SCIENTIFIC BOUNDARIES & LIMITATIONS

RIGOR

1. **Finite Rank:** A matrix $S \in \mathbb{R}^{d \times d}$ has rank $\leq d$; it cannot store arbitrarily many independent associations without interference as T grows relative to d .

2. **Decay Dilemma:** In this recurrence, $\lambda < 1$ prevents unbounded norm growth but creates an exponential forgetting horizon (λ^t).

3. **Quasi-Orthogonality:** Clustered semantic embeddings cause cross-talk earlier than orthogonal codebooks.

4. **Toy Demarcation:** The simulator is an educational toy model, not production BDH.

8. KEY REFERENCES

PRIMARY LITERATURE

1. Kosowski et al. (Pathway, 2025). *The Dragon Hatchling*. arXiv:2509.26507.

2. Engdahl et al. (Pathway, 2026). *BDH-CQ: In-Context Recurrent Latent Reasoning*. arXiv:2608.09888.

3. Schlag, Irie, Schmidhuber (2021). *Linear Transformers Are Fast Weight Programmers*. ICML.

4. Sun et al. (Microsoft, 2023). *Retentive Network (RetNet)*. arXiv:2307.08621.

5. Gu & Dao (2023). *Mamba: Selective State Spaces*. arXiv:2312.00752.

3. WHY IT MATTERS: KV-CACHE PRESSURE

ENGINEERING CONTEXT

In standard Transformers, caching past token representations induces linear growth in cached-token memory ($O(T \cdot d \cdot L)$), leading to GPU VRAM exhaustion, severe memory-bandwidth bottlenecks during autoregressive decoding, and high serving costs on long-horizon reasoning tasks.

4. COMPARATIVE ARCHITECTURAL TRADE-OFFS

TAXONOMY

Architecture	Memory Footprint	Inference Step	Long-Horizon Failure Mode
Transformer	Linear $O(T \cdot d)$	$O(T)$ Memory-bound	VRAM OOM exhaustion
Vector RNN	Fixed $O(d)$	$O(d^2)$ Compute-bound	Bottleneck (d scalar states)
Linear SSM	Fixed $O(d^2)$ in T	$O(d^2)$ Compute-bound	Rank saturation & cross-talk
BDH (Pathway)	Fixed in T for config	Fixed in T (latent)	Synaptic saturation in latent space